

Commands & Options

First, let's get used to *the syntax* of writing *cmdlet's* (commands).

- Commands follow a **Verb-Noun** syntax.

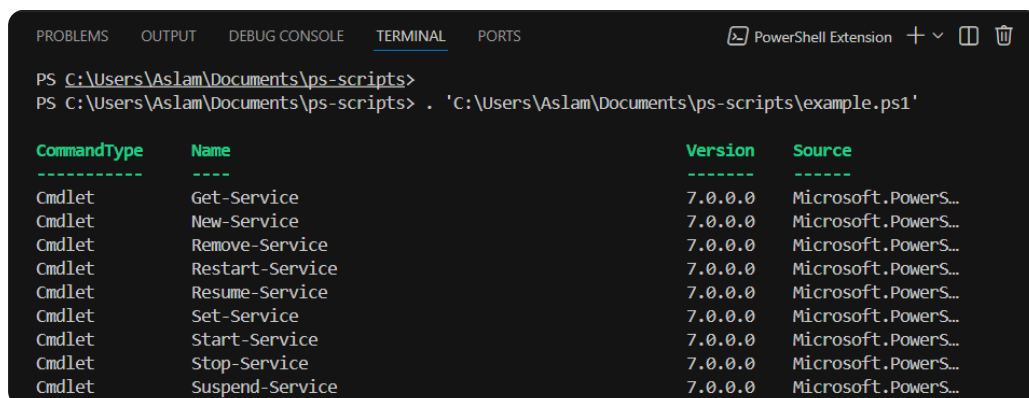
Verb - *Get, Set, Remove, New, Start, Stop, Exit, Join*

Noun - *Service, Process, Item, Content*

Now, you want to do **something relating to a certain noun or verb**, like you want to search for "Service" (*Any running program*) related commands, you know you want to do *something with Services*.

- Use **Get-Help -Verb OR -Noun 'phrase'**

Outputs →



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\Users\Aslam\Documents\ps-scripts>
PS C:\Users\Aslam\Documents\ps-scripts> . 'C:\Users\Aslam\Documents\ps-scripts\example.ps1'
```

CommandType	Name	Version	Source
Cmdlet	Get-Service	7.0.0.0	Microsoft.PowerS...
Cmdlet	New-Service	7.0.0.0	Microsoft.PowerS...
Cmdlet	Remove-Service	7.0.0.0	Microsoft.PowerS...
Cmdlet	Restart-Service	7.0.0.0	Microsoft.PowerS...
Cmdlet	Resume-Service	7.0.0.0	Microsoft.PowerS...
Cmdlet	Set-Service	7.0.0.0	Microsoft.PowerS...
Cmdlet	Start-Service	7.0.0.0	Microsoft.PowerS...
Cmdlet	Stop-Service	7.0.0.0	Microsoft.PowerS...
Cmdlet	Suspend-Service	7.0.0.0	Microsoft.PowerS...

Now that you have *the command you want*, **you'd also** want to understand how to use the command. Simply put that below

Get-Help Get-Service

Outputs →

```
NAME
    Get-Service

SYNTAX
    Get-Service [[-Name] <string[]>] [-DependentServices] [-RequiredServices] [-Include <string[]>]
    [-Exclude <string[]>] [<CommonParameters>]

    Get-Service -DisplayName <string[]> [-DependentServices] [-RequiredServices] [-Include <string[]>]
    [-Exclude <string[]>] [<CommonParameters>]

    Get-Service [-DependentServices] [-RequiredServices] [-Include <string[]>] [-Exclude <string[]>]
    [-InputObject <ServiceController[]>] [<CommonParameters>]

ALIASES
    gsv

REMARKS
    Get-Help cannot find the Help files for this cmdlet on this computer. It is displaying only partial
    help.
    -- To download and install Help files for the module that includes this cmdlet, use Update-Help.
```

In order to actually code in Powershell, you want to actually know useful commands.

Useful commands are normally; COMBINED commands (|), or have **extra arguments**

- Don't worry about the actual commands themselves, try to understand the idea of "|"
-

Let's do a combined command.

```
Get-Service | Where-Object Status -eq "Stopped"
```

Get Service, normally *prints out EVERY service*,

- What the "|" does is take the output of whats on the left, and apply a sort of "filter" on it (*only the stopped ones*).

```
Status      Name      DisplayName
-----
Stopped AarSvc_b220c Agent Activation Runtime_b220c
Stopped ADPSvc Aggregated Data Platform Service
Stopped ALG Application Layer Gateway Service
Stopped AppMgmt Application Management
Stopped AppReadiness App Readiness
Stopped AppVClient Microsoft App-V Client
Stopped ApxSvc Windows Virtual Audio Device Proxy Se...
Stopped AssignedAccessMan... AssignedAccessManager Service
Stopped autotimesvc Cellular Time
Stopped AxInstSV ActiveX Installer (AxInstSV)
Stopped BcastDVRUserServi... GameDVR and Broadcast User Service_b2...
Stopped BEService BattlEye Service
Stopped BITS Background Intelligent Transfer Servi...
Stopped BluetoothUserServ... Bluetooth User Support Service_b220c
Stopped BTAGService Bluetooth Audio Gateway Service
Stopped BthAvctpSvc AVCTP service
Stopped CaptureService_b2... CaptureService_b220c
Stopped CertPropSvc Certificate Propagation
```

We can actually do MORE than one "|".

- Get-Service | Group-Object Status | Select-Object Name, Count

It runs "Get-Service", then *groups all Services by Status*, (Get-Service | Group-Object Status). Then takes the ENTIRE output from the 2 combined left and spits out the Name & Count of each status.

Let us run an actually useful command, *and understand the syntax*.

```
Get-Process | Sort-Object CPU -Descending | Select-Object -First 10 Name,  
CPU, Id
```

*Get-Process shows every process, Sort-Object CPU -Descending takes the CPU attribute that Get-Process has, and sorts it in **Descending Order**.*

*Select-Object -First 10 Grabs the **first 10 processes** sorted in Descending order. Name, CPU, Id only **spits out those columns** for each Object.*

- Here's what mine outputs →

Name	CPU	Id
----	---	--
Discord	1677.97	14688
audiodg	489.33	4380
Discord	483.67	11888
Discord	416.42	3716
zen	400.38	18812
zen	365.48	14212
zen	321.14	15064
zen	176.69	21996
zen	175.95	19204
Discord	147.80	11772

What's **useful about this command**; is because, certain virus's can hide from Task Manager, this command can find them and kill them (*Stop-Process -Id 14688*)

Next PDF will show logical programming (Variables & For Loops), and more useful commands alongside their syntax's.